Department of Electronic & Telecommunication Engineering, University of Moratuwa, Sri Lanka.

Design and Implementation of a Fully Analog Control System for DC Motor Speed Control

Team Neural Nexus – Group No. 12

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1 Abstraction of methodology

The motor speed control system is designed as a fully-analog closed-loop control system. The process begins with an encoder channel from the motor, which feeds into a frequency-to-voltage converter. This produces an encoder signal voltage scalar ranging from 0 to 12V, providing a proportional voltage signal for the motor speed. This feedback signal is compared against a set reference speed to calculate the error. The error signal is processed by a PID controller to generate a control signal. Concurrently, a triangular waveform generator produces a carrier wave. The control signal and the triangular waveform are compared to generate a PWM signal, which is then amplified using a MOSFET-based PWM amplifier to drive the motor.

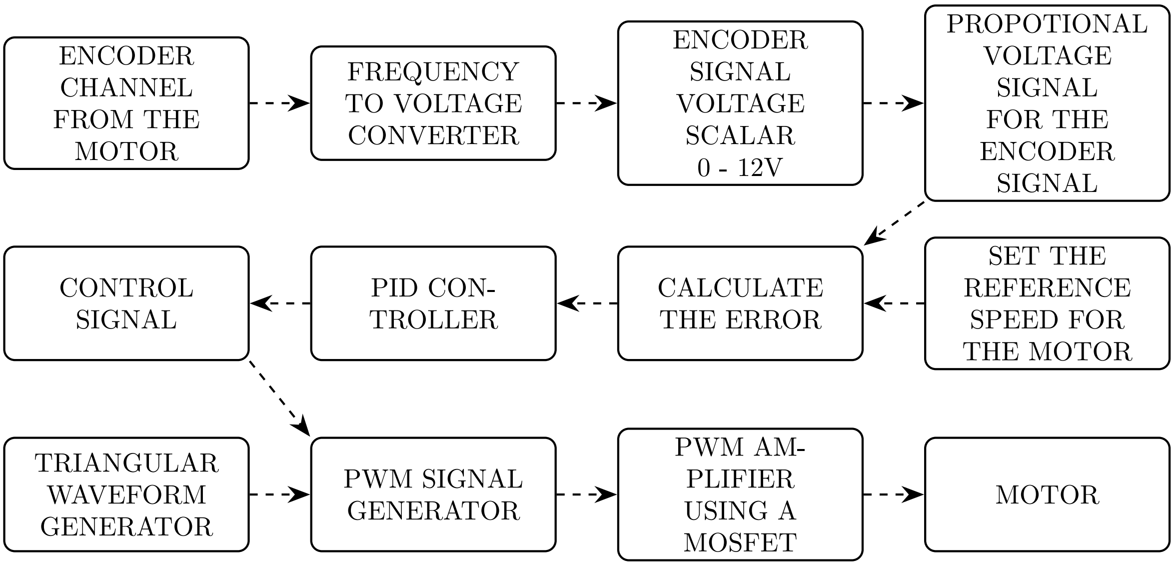


Figure 1: Flow diagram of the DC motor speed control system

2 Full schematic

The full schematic integrates the core functional subsystems of the analog control system: the frequency-to-voltage converter, the PID controller, the PWM signal generator, and the PWM amplifier. These interconnected analog circuits work together sequentially to form the complete closed-loop feedback system on the final breadboard implementation.

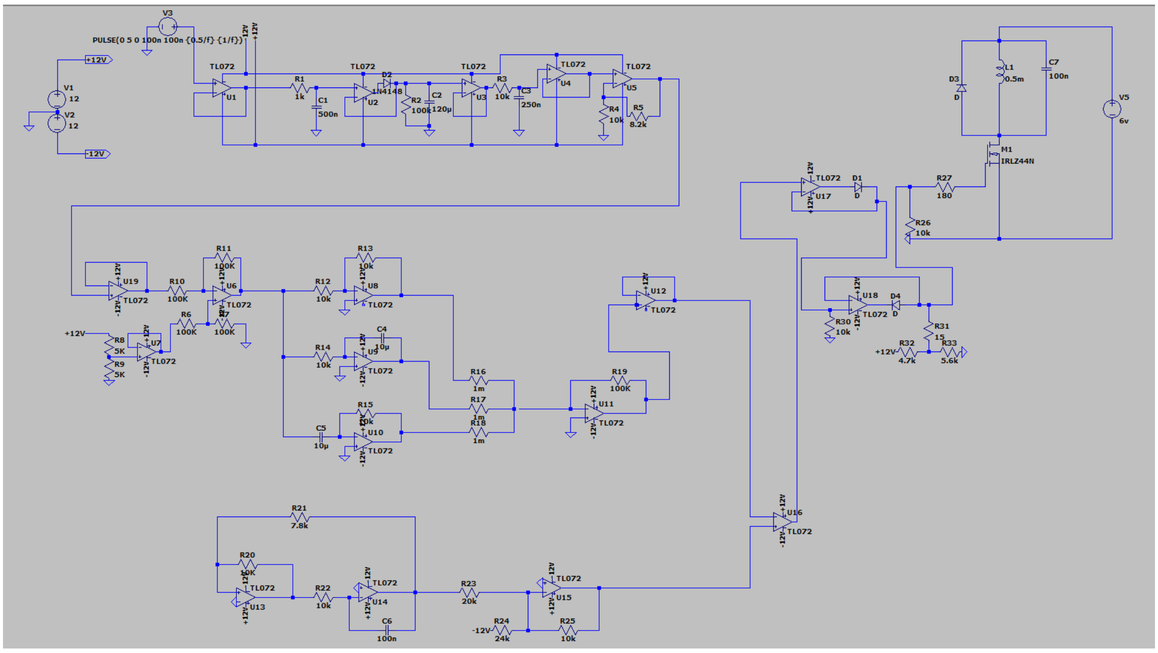


Figure 2: Complete circuit schematic

3 Main functional modules

The system comprises the following main functional modules to achieve fully-analog speed control:

- Frequency to Voltage Converter
- Error Calculation and PID Controller
- Triangular Waveform and PWM Signal Generator
- PWM Amplifier (MOSFET Driver Stage)

4 PWM Signal Generation

4.1 Schematic Design

Figure 8 shows the complete circuit schematic of the PWM signal generation system.

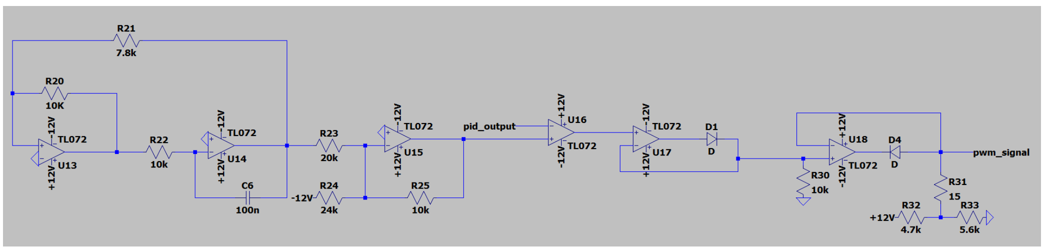


Figure 3: Complete circuit schematic of the PWM signal generation system.

4.1.1 Triangular Wave Generation

In the Analog motor speed controller, pulse-width modulation (PWM) is used to regulate the speed of the DC motor according to the control voltage generated by the feedback system. A stable triangular carrier waveform is required for PWM generation. This triangular wave is produced using a closed-loop analog oscillator consisting of a non-inverting Schmitt trigger and a ramp generator (integrator). The circuit is implemented entirely using analog components and does not require an external clock source.

The Schmitt trigger generates a square wave whose output alternates between positive and negative saturation levels. This square wave is then fed to a ramp generator to integrate it and produce a triangular waveform. The generated triangular wave is fed back to the Schmitt trigger, forming a closed-loop oscillation system. The frequency and amplitude of the triangular wave are determined by the resistor and capacitor values used in the circuit.

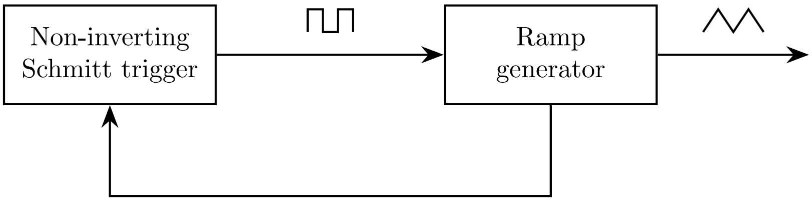


Figure 4: Block diagram of the triangular wave generator using a non-inverting Schmitt trigger and ramp generator

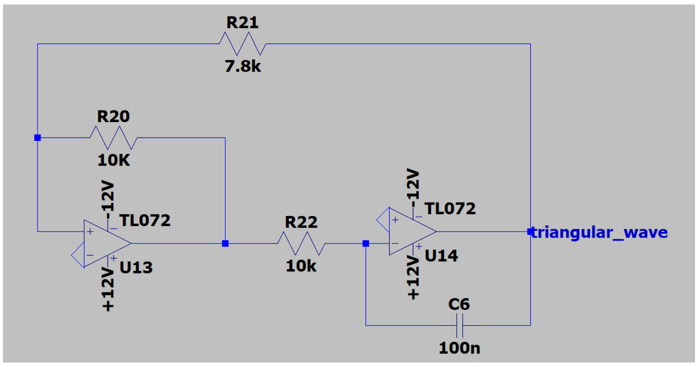


Figure 5: Triangular wave generator circuit.

Figure 6: Triangular waveform.

Oscillation Frequency

The triangular wave is generated using an op-amp integrator. The rate at which the output ramps up or down depends on the switching voltage from the Schmitt trigger and the RC time constant of the integrator. For this circuit, the slope of the ramp can be written as,

$$\frac{dV}{dt} = \frac{V_{\text{sat}}}{R_{22}C_6}$$

Here, R_{22} and C_6 form the integrator network, and V_{sat} is the output saturation voltage of the Schmitt trigger.

The capacitor voltage continuously moves between the upper and lower switching levels of the Schmitt trigger, V_{TH+} and V_{TH-}. The time taken for one transition between these two levels is:

$$t = \frac{(V_{TH+} - V_{TH-})R_{22}C_6}{V_{\text{sat}}}$$

One full cycle includes both the rising and falling ramps, so the total period becomes:

$$T = 2 \cdot \frac{(V_{TH+} - V_{TH-})R_{22}C_6}{V_{\text{sat}}}$$

Therefore, the oscillation frequency is:

$$f = \frac{V_{\text{sat}}}{2R_{22}C_6(V_{TH+} - V_{TH-})}$$

4.1.2 Triangular Wave Attenuation and Level Shifting

The PWM signal is generated by comparing the triangular carrier waveform with the output of the PID controller. The PID output voltage determines the required motor speed, where positive voltages correspond to forward motion. Since the proposed speed control system is designed to operate only in the forward direction, negative PID outputs are not used to drive the motor.

In the original control strategy, a positive PID output would command forward motion, a zero output would stop the motor, and a negative output would command reverse motion. However, reverse motor operation and the associated direction sensing circuitry were omitted in this design because they require digital circuits. So, whenever the PID output becomes negative, the motor is set to remain stationary.

To realize this behavior using PWM, the triangular carrier waveform is attenuated and level shifted so that its voltage range matches only the positive operating range of the PID output. As a result, PWM pulses are generated only when the PID output is positive. When the PID output falls below zero, no intersection occurs between the PID signal and the shifted triangular waveform, resulting in a zero duty cycle and hence stopping the motor.

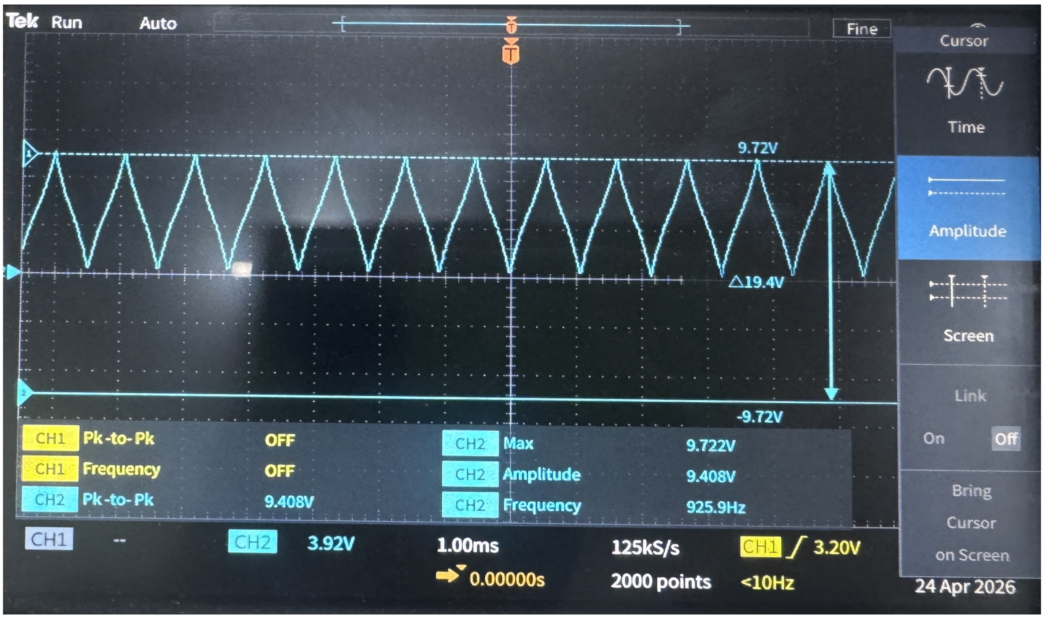
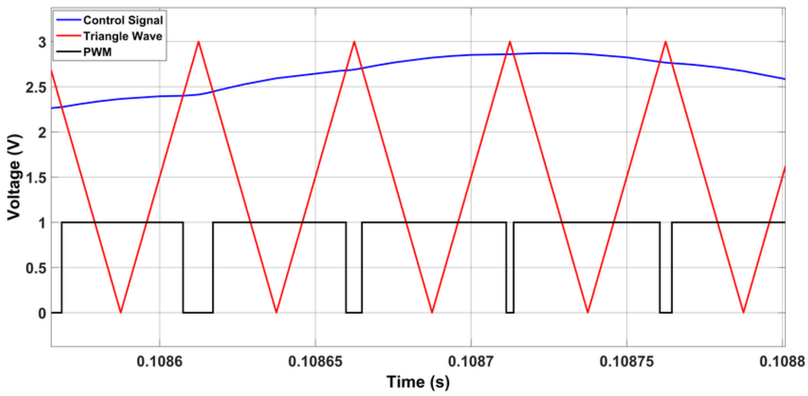


Figure 7: Shifted Triangular waveform.

4.1.3 PWM Generation

The PWM signal is generated by comparing the level-shifted triangular carrier waveform with the output of the PID controller using a voltage comparator. Whenever the PID control voltage exceeds the instantaneous value of the triangular waveform, the comparator output switches to its positive saturation level; otherwise, it switches to its negative saturation level. Consequently, the duty cycle of the PWM signal varies in accordance with the PID output voltage, thereby controlling the average voltage applied to the motor.

Since the comparator is powered using a dual supply, the generated PWM waveform swings between approximately (+12 V) and (−12 V). However, the subsequent stages of the motor drive circuit require a unipolar control signal. Therefore, the bipolar PWM signal is conditioned in the following stages to obtain a (0 V) to (5 V) PWM signal. This signal is then used to control the analog motor driver stage.



Example PWM generation by comparing a PID control signal with a triangular waveform

4.1.4 Precision Rectifier

The precision rectifier is used to remove the negative portion of the bipolar PWM signal while preserving the positive pulses. Consequently, the output of this stage is a unipolar PWM waveform ranging from (0 V) to (+12 V). The use of a precision rectifier ensures accurate clipping without the voltage drop associated with ordinary diode circuits.

4.1.5 Positive Clipper

The output of the precision rectifier still has an amplitude of approximately (0 V) to (+12 V). Since the subsequent motor drive stage operates with a maximum control voltage of (5 V), a positive clipper circuit is employed to limit the amplitude of the PWM signal. The circuit clips all voltages above (5 V), producing a PWM waveform in the range of (0 V) to (5 V) while preserving the duty cycle of the signal.

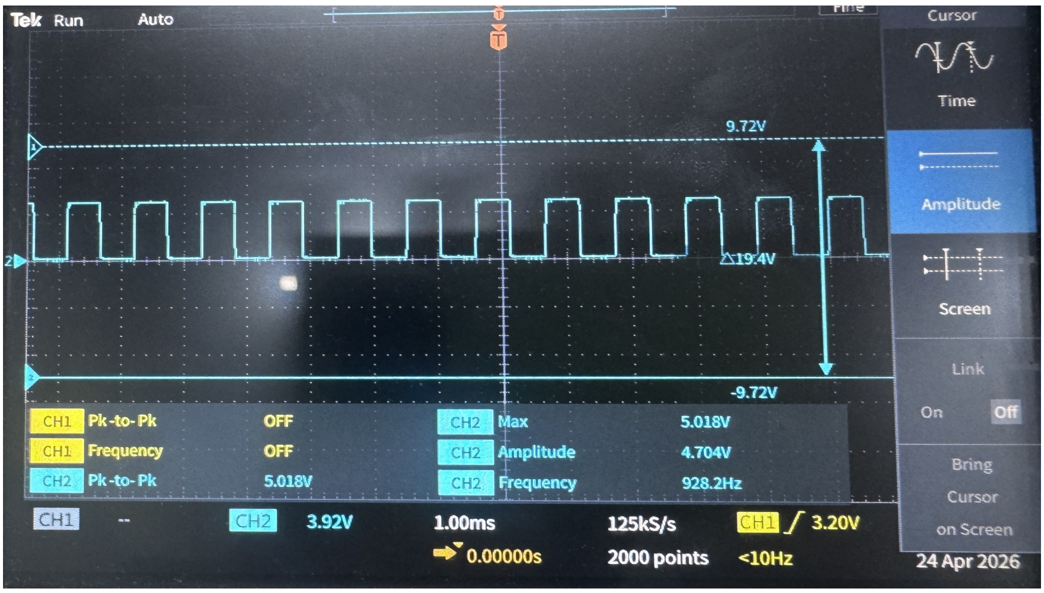


Figure 8: Final PWM signal.

4.2 opAmp Selection

The TL072 operational amplifier was selected for implementing the PWM generation stages due to its high slew rate of 13 V/µs, compared to other commonly available operational amplifiers in Sri Lanka. Since the PWM generation process involves rapid switching between voltage levels, a high slew rate is essential to produce sharp rising and falling edges in the PWM waveform. Sharper edges improve the accuracy of the PWM signal and reduce distortion in the subsequent analog processing stages. Therefore, the TL072 was chosen to ensure reliable and high-quality PWM signal generation.

5 PWM amplifier

5.1 Schematic Design

The PWM amplifier circuit is engineered around an N-Channel MOSFET to drive the motor using the PWM signal generated by the preceding stages. It consists of necessary current-limiting elements, including a gate resistor and a gate pull-down resistor, along with a Schottky flyback diode for inductive kickback protection, and a decoupling capacitor designed for high-frequency EMI noise filtering.

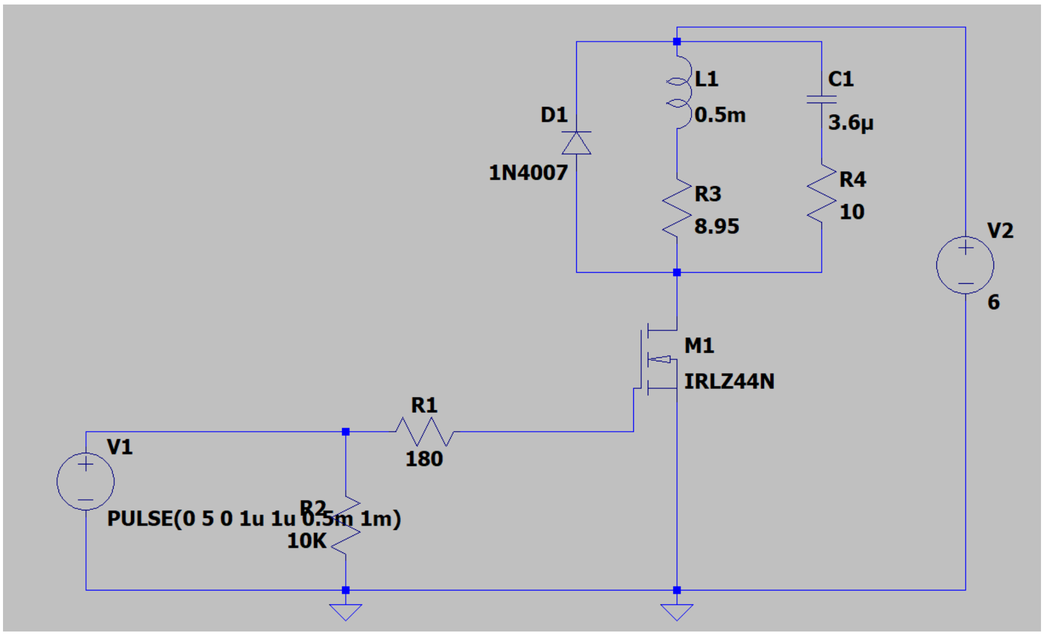


Figure 9: Schematic of PWM amplifier

5.2 Simulation or lab test results

Simulation results validate that the PWM amplifier successfully drives the motor load. The analysis of the output waveform demonstrates the efficiency of the chosen switching frequency. Furthermore, it verifies the effectiveness of the flyback diode, which safely loops the collapsing magnetic energy and successfully clamps the inductive spikes (maximum drain voltage) to approximately 6.4V.

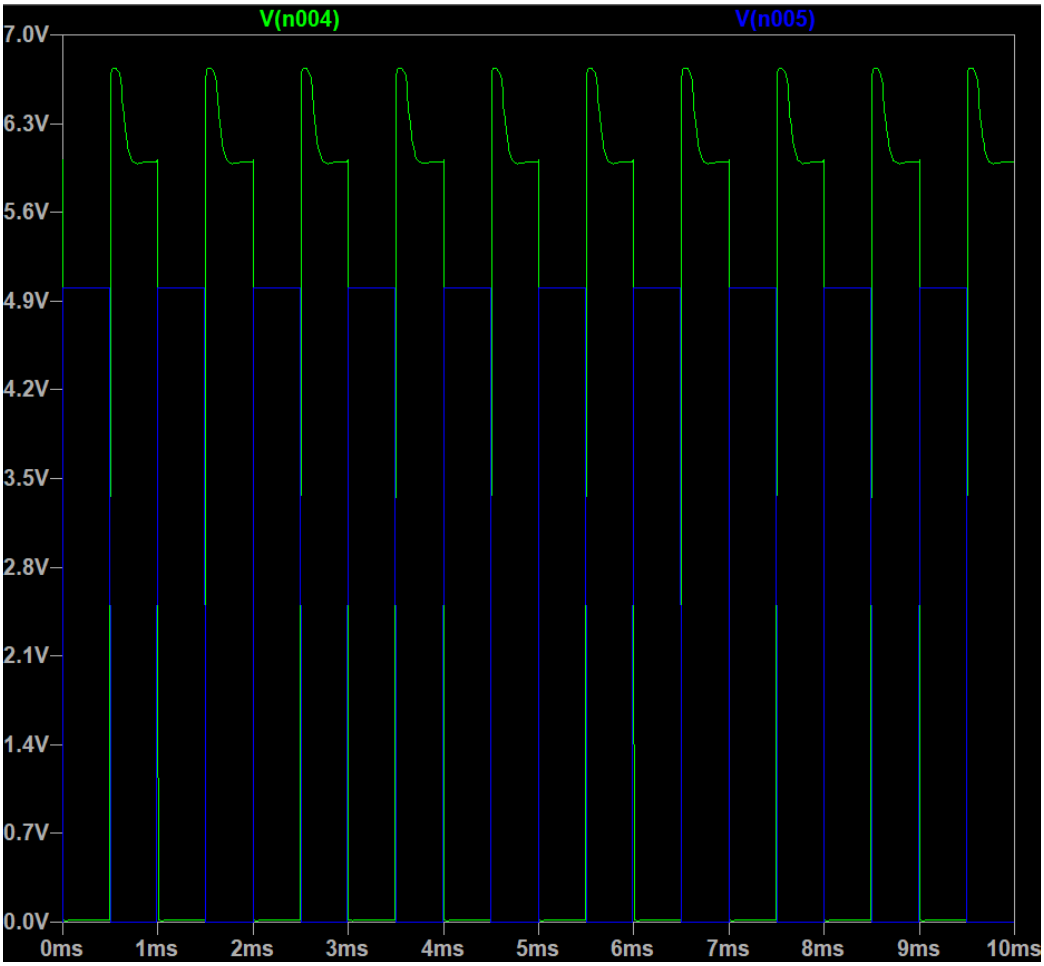


Figure 10: Blue : 5V PWM , Green : 6V amplifed PWM, the spike in the simulation is due to the 1n4007 diode using the practical diode(1N5819 Schottky Flyback Diode) we used, was not available in ltspice and it also couldnot be modelled as well

5.3 Component Selection

5.3.1 IRLZ44N (N-Channel MOSFET)

- **Role:** Primary low-side electronic switch.
- **Justification:** Logic-level threshold ($V_{GS(th)} = 1.0\text{V} - 2.0\text{V}$) ensures full saturation driven by a 5V PWM, minimizing $R_{DS(on)}$ to 0.022 Ω for highly efficient switching. Its 47A rating handles the 0.67A stall current with zero need for a heatsink.

5.3.2 180 Ω (Gate Resistor)

- **Role:** Inrush current limiter for the driving source.
- **Justification:** Caps the peak draw from the op-amp/MCU to a safe ≈ 28 mA ($I = 5\text{V}/180\ \Omega$). It maintains an incredibly fast switching time of ≈ 1.53 μs, which is negligible against the 1 ms period of the 1 kHz PWM signal. The RC time constant (τ) determines the switching speed:

$\tau = R \cdot C_{iss} = 180 \cdot (1700 \times 10^{-12}) = 306$ ns. It takes about 5τ to fully charge the gate (≈ 1.53 μs).

5.3.3 10 kΩ (Gate Pull-Down Resistor)

- **Role:** Safety bleed resistor to prevent a floating gate.
- **Justification:** Guarantees the motor stays OFF if the PWM source disconnects. Forms a voltage divider with R1 that still delivers 4.91V to the gate during active HIGH states, keeping the MOSFET fully saturated while wasting only 0.5 mA.

5.3.4 1N5819 (Schottky Flyback Diode)

- **Role:** Inductive kickback clamper. (Very short reverse recovery time)
- **Justification:** Safely loops the collapsing magnetic energy, clamping the maximum drain voltage to $\approx$ 6.4V. The Schottky's near-zero reverse recovery time (t_{rr}) prevents destructive shoot-through currents during the 1 kHz switching cycles, outperforming standard silicon rectifiers.

5.3.5 100 nF (Decoupling Capacitor)

- **Role:** High-frequency EMI / brush noise filter.
- **Justification:** Acts as a frequency-dependent short circuit. At the 1 kHz driving frequency, it has high impedance ($\approx$ 1.5 kΩ) and ignores the PWM. At the 10 MHz noise frequencies generated by physical brush arcing, it drops to $\approx$ 0.16 Ω, safely shunting EMI away from sensitive analog control stages.

6 Frequency to Voltage converter

6.1 Working Principle

The encoder generates a constant-amplitude pulse train whose frequency is directly proportional to the motor speed. This frequency is converted into a proportional DC voltage using an analog signal conditioning circuit consisting of an RC low-pass filter followed by a peak detection stage.

The system operates based on the following principle:

- Higher rotational speed → higher pulse frequency
- Higher frequency → higher effective output voltage
- Result: DC voltage proportional to motor speed

6.2 Signal Processing Stages

6.2.1 RC Low-Pass Filter

The pulse train from the encoder is passed through an RC low-pass filter. This converts the square wave into a smoothed waveform whose amplitude depends on the input frequency.

The cutoff frequency is given by:

$$f_c = \frac{1}{2\pi RC}$$

Behavior:

- At low frequencies: capacitor fully charges and discharges
- At high frequencies: capacitor does not fully discharge, leading to higher average voltage

6.2.2 Peak Detector Circuit

A diode-capacitor peak detector is used to capture and hold the peak voltage of the filtered signal.

Operation:

- Fast charging occurs through the diode when input rises
- Slow discharge occurs through the resistor when input falls

This results in a quasi-DC voltage proportional to the signal frequency.

6.3 Dynamic Behavior

The circuit exhibits asymmetric charging and discharging characteristics:

- **Fast rise time:** Capacitor charges rapidly through the diode (low forward resistance)
- **Slow fall time:** Capacitor discharges slowly through the resistor network

This behavior ensures that the output voltage closely follows the envelope of the input frequency variations.

6.4 Advantages of the Implementation

- Simple and low-cost analog implementation
- No requirement for dedicated frequency-to-voltage conversion ICs
- Suitable for embedded systems with limited processing resources
- Real-time response without digital computation delay

6.5 Limitations

- Reduced accuracy at low frequencies due to poor averaging
- Sensitive to noise and ripple in the output signal
- Strong dependence on RC time constant selection
- Requires careful tuning for stable operation across the speed range

6.6 Component Selection

6.6.1 Operational Amplifier: TL072

The TL072 JFET-input operational amplifier is used due to:

- High slew rate suitable for fast signal transitions
- Wide bandwidth for handling encoder frequencies up to approximately 1.7 kHz
- Very low input bias current, minimizing loading effects on the RC network

6.6.2 Diode: 1N4148

The 1N4148 is selected for the peak detector stage because:

- Very fast switching speed
- Low junction capacitance
- Suitable for high-frequency pulse signals from the encoder

6.7 Schematic Design

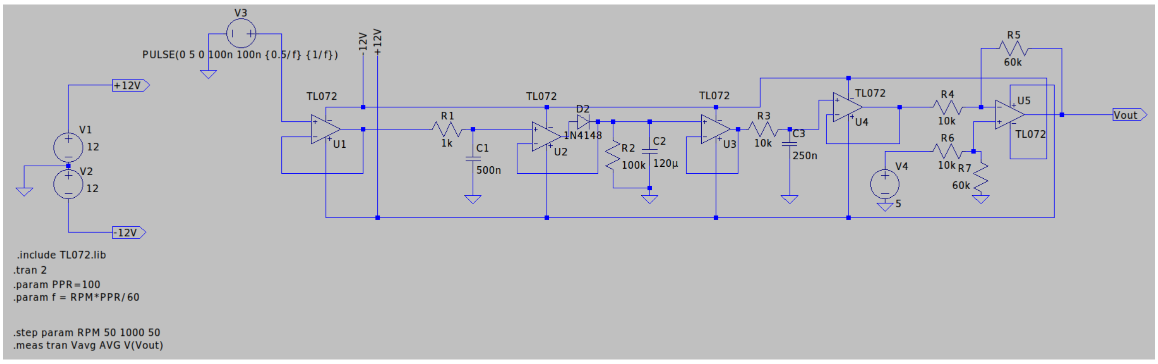


Figure 11: Frequency to Voltage Converter

6.7.1 Simulation or lab test results

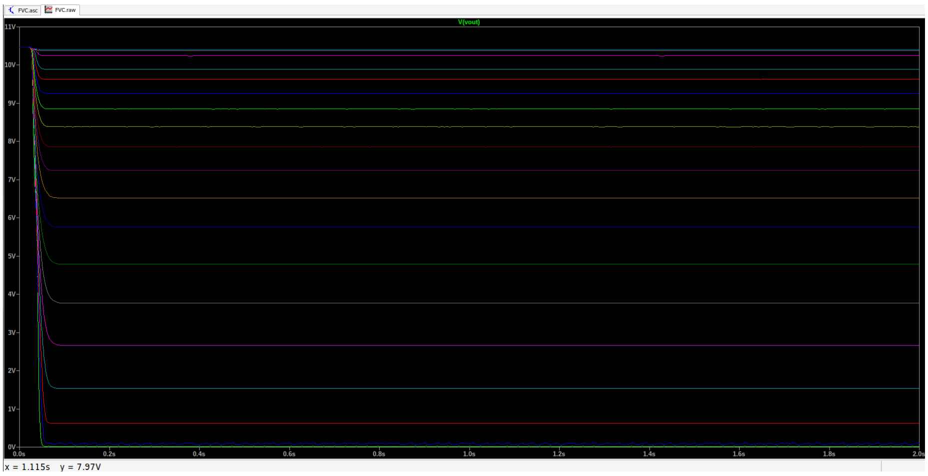


Figure 12: Frequency to Voltage Converter Simulation

7 PID Controller Design and Experimental Results

7.1 PID Control Equation

The output of a Proportional–Integral–Derivative (PID) controller is given by

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau)\, d\tau + K_D \frac{de(t)}{dt}, \tag{1}$$

where

$$e(t) = r(t) - y(t). \tag{2}$$

Here, $r(t)$ is the required set-point, $y(t)$ is the measured feedback signal, and $e(t)$ is the error between the set-point and the measured output. The terms K_P, K_I, and K_D represent the proportional, integral, and derivative gains, respectively.

- **Proportional action:** The proportional term deals with the present value of the error. It determines how strongly the controller reacts to the difference between the set-point and the measured output.
- **Integral action:** The integral term accumulates the error over time. The main responsibility of the integral action is to remove the steady-state error.
- **Derivative action:** The derivative term responds to the rate of change of the error. We use the derivative controller to react to sudden changes in the set-point.

7.2 Schematic Design

Figure 13 shows the schematic design of the PID-based motor-control system. The required motor speed is represented by a voltage signal, which can be adjusted using a variable resistor. We take this input and the current frequency-to-voltage converter output and compare them to generate the error signal.

This error signal is then fed to the PID controller. The output represents the required control effort needed to make the motor speed follow the set-point.

After passing through the PID controller, the output is compared with the triangular waveform generator output to generate a PWM signal.

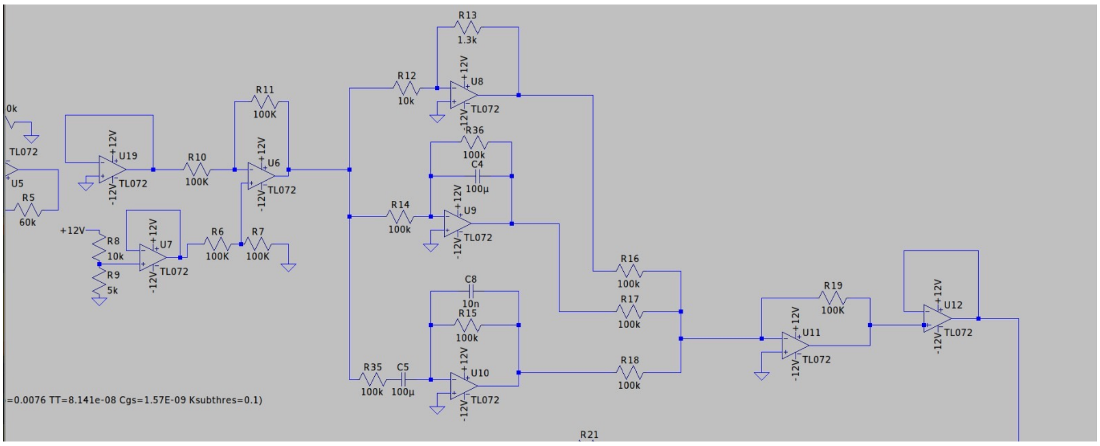


Figure 13: Schematic design of the PID-based motor-control system.

7.3 Effect of Proportional Gain

We can observe that, in the region where K_P was low, there were large variations in the output motor speed. The main reason is that the proportional action was not sufficient to control the output, so the error remained until it became considerably large enough for the controller to act.

When K_P was increased, we could observe that the variations became visibly smaller. This is because even a small offset voltage produced a considerable error signal through the proportional action, resulting in immediate correction.

However, K_P should not be increased to excessively high values, as this may cause oscillations, overshoot, or instability in the motor-control system.

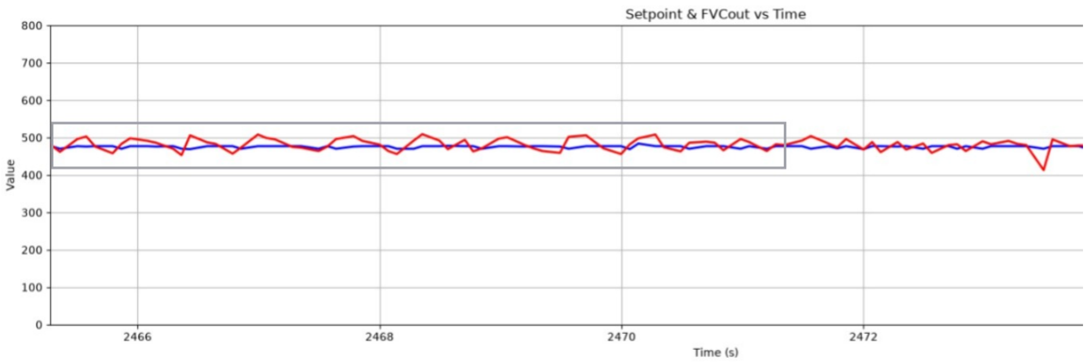


Figure 14: Experimental result showing the effect of proportional gain.

7.4 Effect of Derivative Gain

When K_D was low, we could observe that there was a visible lag in the motor speed compared with changes in the set-point. The main reason is that, since K_D was low, the responsiveness to changes was also low. Therefore, the output motor speed waited until a large change occurred.

When K_D was increased, we could observe that the lag was reduced and that the motor speed smoothly followed the set-point voltage. This is because the controller could quickly respond to any change that occurred and adapt to it.

However, an excessively high derivative gain should be avoided because the derivative action can amplify high-frequency noise in the feedback signal.

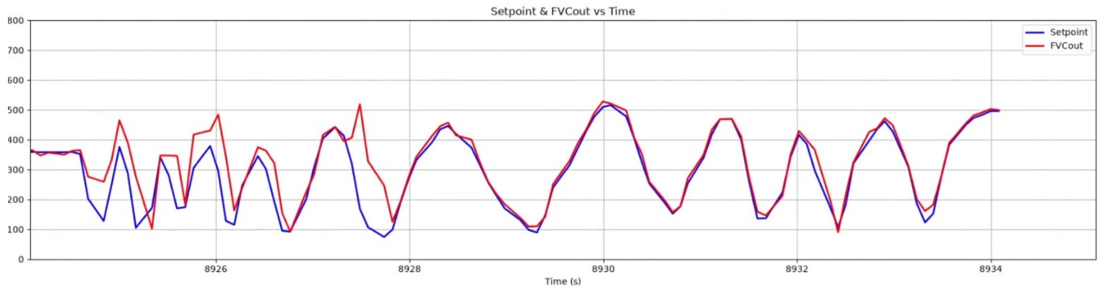


Figure 15: Experimental result showing the effect of derivative gain.

7.5 Effect of Integral Gain

Figure 16 shows the system response when the integral gain was low. In particular, when K_I was set to a very small value, a steady-state error was visible between the set-point and the output motor speed.

Using the integral term, we accumulate the error over time, but a low value of K_I produced insufficient accumulated correction.

When K_I was gradually increased, the correction was sufficient to remove the steady-state error.

However, an excessively high integral gain may cause overshoot, oscillations, and integral windup. Therefore, K_I must be selected carefully to remove the steady-state error while maintaining a stable motor response.

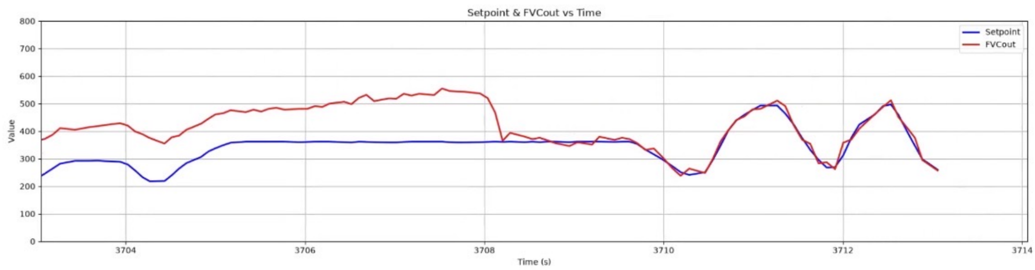


Figure 16: Experimental result showing the effect of integral gain.

8 Conclusion

This project successfully demonstrated the design, implementation, and tuning of a fully-analog closed-loop control system for DC motor speed regulation. By utilizing fundamental analog electronics rather than digital microcontrollers, the system effectively achieved real-time, continuous feedback control. The feedback loop was established using a custom Frequency-to-Voltage Converter (FVC) that accurately translated motor encoder pulses into a proportional DC voltage. This enabled the analog PID controller to continuously compute the error against a set reference voltage and dynamically adjust the control effort based on the tuned gains.

Experimental validation confirmed the theoretical behavior of the PID controller: the proportional (K_P) action provided immediate correction for gross speed variations, the integral (K_I) action successfully eliminated steady-state error, and the derivative (K_D) action improved overall system responsiveness while mitigating lag. Additionally, the custom PWM generation circuit, built around a stable triangular waveform oscillator and TL072 operational amplifiers, proved highly capable of translating the analog PID control voltage into a precise unipolar PWM signal. Finally, the IRLZ44N MOSFET amplifier stage reliably drove the motor load, safeguarded by a Schottky flyback diode to suppress inductive spikes. Ultimately, the successful breadboard implementation validated that precise, robust, and stable motor speed control can be achieved entirely through an analog hardware architecture.